School of Information and Technology Engineering
Addis Ababa Institute of Technology
Addis Ababa University

Maleware Analysis individual Assignment

Hanamariam Yehuala–UGR/9409/13

To: Tamrat Aweta

Malware Analysis Report

Sample Name: sample.exe
Analyst: Hanamariam Yehuala
Tools Used: Detect It Easy (DIE), PEview, Process Monitor (Procmon), Wireshark, Regshot
Environment: FlareVM

1. Executive Summary

This report analyzes `sample.exe`, an obfuscated 32-bit .NET executable exhibiting highly intrusive behavior. It performs over 44,000 registry changes, modifies the file system, and conducts basic network checks. This extensive system interaction suggests a deep integration attempt and likely persistence objective. Network activity (ICMP pings) indicates readiness for Command and Control (C2). `sample.exe`'s nature points to a malicious purpose aimed at system compromise.

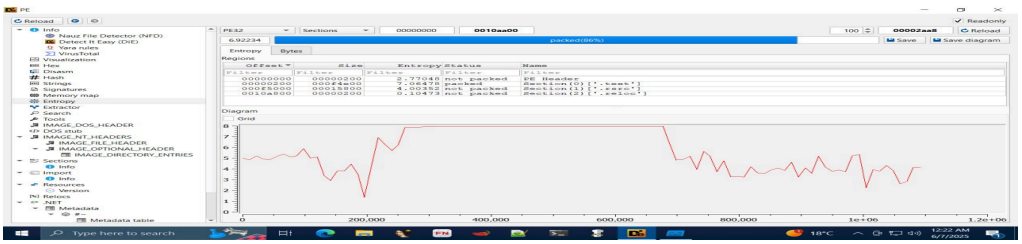
2. Brief Background

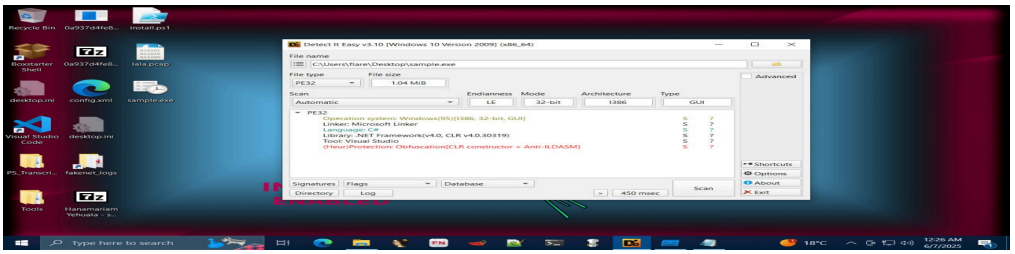
`sample.exe` underwent static and dynamic analysis to understand its malicious capabilities. Its inherent obfuscation, intrusive system modifications, and network outreach collectively confirm it as a non-benign application, likely focused on persistence, information gathering, and further compromise.

3. Technical Behavior Analysis

3.1. Static Analysis

Initial static analysis identified `sample.exe` as a PE32 (32-bit) .NET executable, approximately 1.04 MB. A critical finding was its obfuscation/packing (Anti-ILDASM), indicated by high entropy (6.92) across 86% of the file. This strongly suggests an intent to evade detection and complicate reverse engineering.





3.2. Dynamic Analysis

Dynamic analysis details the behavior of sample.exe during execution in a controlled environment.

3.2.1. Execution Flow

Upon execution, **`sample.exe`** initiated a new process (PID 3000) and immediately began loading numerous critical **Dynamic-Link Libraries (DLLs)** essential for its operation and interaction with the Windows operating system. These include fundamental system DLLs such as C:\Windows\System32\ntdll.dll, C:\Windows\System32\wow64win.dll, C:\Windows\System32\wow64cpu.dll, C:\Windows\System32\wow64log.dll, and C:\Windows\System32\user32.dll. Additionally, it loaded various .NET Framework DLLs (e.g., `mscoree.dll`, `clr.dll`), confirming its .NET dependency from static analysis. This initial phase serves as the malware's preparation for subsequent intrusive actions.

.3.2.2. Registry Interactions

`sample.exe` exhibits an **extraordinary interaction with the Windows Registry**, logging a staggering **44,557 total changes**. This includes **20,315 new keys and 23,957 new values added**, with 212 values modified and no deletions.

This massive footprint suggests a deliberate effort to deeply embed itself within the system, likely for configuration, operational parameters, or stealthy integration. Key areas of activity include:

- **User-Specific Hives (`HKU\S-1-5-21-...\`):** Extensive changes within the current user's `Explorer` settings, potentially altering user experience, hiding files, or establishing subtle persistence.
- **Identity-Related Keys (`HKLM\SOFTWARE\Microsoft\IdentityCRL\`):** Modifications to components involved in identity and authentication, raising concerns about potential credential theft, session hijacking, or authentication manipulation.

The overwhelming volume of additions, without deletions, indicates `sample.exe` is primarily adding its own components or configurations to the system.

3.2.3. File System Operations

`sample.exe` creates files in `C:\Windows\`, a common malware tactic for dropping components, achieving persistence, or evading detection. It also queries file attributes and uses `CreateFileMap`/`MapViewOfSection`, suggesting shared memory use or code injection.

3.2.4. Persistence Method

While no single `Run` key was observed, the massive registry additions (over 44,000 changes) and file creation in `C:\Windows\` strongly suggest a sophisticated, multi-point persistence strategy, likely utilizing obscure startup locations.

3.2.5. Command and Control (C2)

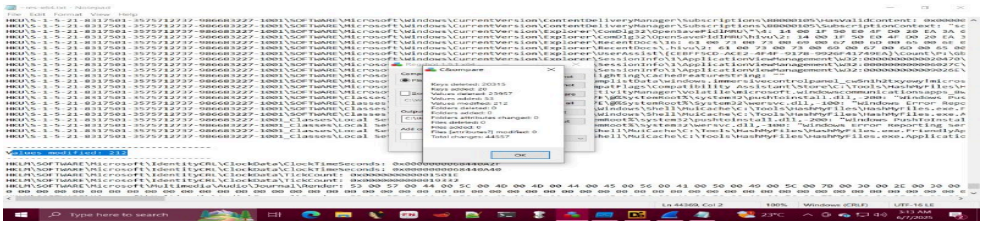
Network activity shows `sample.exe` performing basic connectivity checks via **ICMP pings to 8.8.8.8**. This confirms internet reachability, preparing the malware for potential C2 communication, module downloads, or data exfiltration, even if complex C2 protocols weren't observed in the snippet.

4. Indicators of Compromise (IOCs)

Based on analysis, key IOCs include:

- **File Name:** `sample.exe`
- **Process ID:** 3000
- **Network Connection:** ICMP to `8.8.8.8`
- **Static Indicators:** PE32 (x86), .NET (CLR v4.0.30319), Obfuscation/Anti-ILDASM, High Entropy (approx. 6.92).
- **Registry Modification Footprint:** Abnormally high changes (44,557 total; 20,315 keys, 23,957 values added), particularly in `HKU\S-1-5-21-...\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\` and `HKLM\SOFTWARE\Microsoft\IdentityCRL\`.
- **File Creation Location:** `C:\Windows\` directory.

5. **Detection and Mitigation Challenges**

sample.exe presents significant detection and mitigation challenges due to:

- **Obfuscation/Packing:** Hinders static analysis and signature-based detection.
- **High Volume of Registry Changes:** Over 44,000 modifications obscure malicious entries and cause alert fatigue.
- **Legitimate API Abuse:** Uses standard Windows APIs, making malicious activity hard to distinguish without advanced analytics.
- **Stealthy Persistence:** Extensive registry changes enable persistence in less monitored locations.
- **Basic C2 Channel:** Initial ICMP pings are hard to block; actual C2 may be obfuscated.
- **Remediation Complexity:** Manual cleanup of numerous, granular registry changes is impractical, requiring precise automation.
- **Potential Anti-Analysis:** Evidence suggests evasion of analysis environments, complicating forensics.